LU_CYCLAM_DFT · README

Coordination of Lu(III) in the Lu–cyclam complex

Code, structures and results of the independent DFT check in the Supporting Information (Section S1). It tests whether [Lu(cyclam)$(H_2O)_5]^{3+}$ is nona-coordinate or octa-coordinate with the fifth water molecule in the second coordination sphere.

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GFN2-xTB + PBE0-D3(BJ)/def2-SVP | Python 3.13 · fixed versions | Reproducibility verified 9 Nov 2026 | MATLAB interface

6.9 kcal/mol

MAIN RESULT

The octa-coordinate [Lu(cyclam)$(H_2O)_4]^{3+}\cdot H_2O$ is the lowest-energy structure; the most stable nona-coordinate isomer lies higher by 6.9 kcal/mol (PBE0-D3(BJ)/def2-SVP, ddCOSMO water).

Nona-coordinate isomer – a structure of the same composition, [Lu(cyclam)$(H_2O)_5]^{3+}$, in which Lu(III) is bonded to nine donor atoms: the four nitrogen atoms of cyclam and the oxygen atoms of all five water molecules (coordination number 9; 4N + 5O). It is an isomer of the octa-coordinate form because both contain the same atoms; they differ only in where one water molecule sits – bound to Lu, or hydrogen-bonded in the second coordination sphere (4N + 4O + H_2O). In the code, a donor atom counts as bound when its distance from Lu is below 3.0 Å.

Contents

PATH	CONTENT
`results/xtb_structures/`	All GFN2-xTB structures (.xyz, Å; comment line holds the GFN2-xTB energy in eV)
`results/dft/dft.jsonl`	DFT results, one line per structure (energies in Hartree)
`scripts/`	Python pipeline, steps 01–08 (below)
`matlab/`	MATLAB functions: results table and the published figures
`figures/`	Published Figs. S1 and S2 (from `matlab/lu_dft_figures.m`), PDF and PNG
`results/verification/`	Output of the clean-environment reproducibility check (below)
`requirements.txt`	Exact package versions used
`README_sr.md`	Instructions in Serbian
`SHA256SUMS.txt`	Checksums of all files

Model and input

No experimental data enter the calculation. The input is the chemical model of the cation: one Lu^{3+}, one cyclam (SMILES in `scripts/lib.py`) and five water molecules, total charge +3. Counter-ions (BF_4^-, Na^+) are not included; water is treated as a continuum (ε = 78.36). All starting geometries are generated by the code; random numbers use fixed seeds.

Protocol

STEP	SCRIPT	WHAT IT DOES
1	`01_conformers.py`	16 cyclam conformers (RDKit, MMFF94) × 3 water distributions (5+0, 4+1, 3+2 inner + outer) = 48 structures; GFN2-xTB optimization in ALPB water (f_max < 0.03 eV/Å)
2	`02_rank_xtb.py`	Ranks all structures by GFN2-xTB energy (`xtb_ranking.json`)
3	`03_cn9_cages.py`	Restrained 4N + 5O cages from the best structures, then unrestrained relaxation
4	`04_calibration_aqua.py`	Reference system: $[Lu(H_2O)_9]^{3+}$ vs $[Lu(H_2O)_8]^{3+}\cdot H_2O$
5	`05_dft_single_point.py`	PBE0/def2-SVP, ECP28MWB on Lu, density fitting, ddCOSMO water, + D3(BJ); Mulliken charge on Lu and dipole moments
6	`06_summary.py`	Table of relative energies
7	`07_compare.py`	Compares a new `dft.jsonl` with the deposited one
8	`08_figures.py`	Python version of Figs. S1–S2 (the published figures come from `matlab/lu_dft_figures.m`)

`run_all.py` runs steps 1–6 in order.

Requirements

- Linux or macOS (on Windows: WSL2). PySCF does not run natively on Windows.
- Python 3.10–3.13 (results were produced with Python 3.13 on Linux x86_64), ≥ 8 GB RAM.

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Reproducing the results

All commands from the `scripts/` folder.

A. Inspect the deposited results (seconds)

```
python 06_summary.py ../results/dft/dft.jsonl
python 08_figures.py
```

B. Recompute the DFT energies from the deposited geometries (~1 h 50 min on 2 cores)

```
python run_all.py --skip-xtb --threads 4
python 07_compare.py ../work/dft.jsonl
```

C. Full pipeline including the conformer search (~2 h on 2 cores)

```
python run_all.py --threads 4 --jobs 4
python 07_compare.py ../work/dft.jsonl
```

`07_compare.py` reports the difference from the deposited energies in kcal/mol and exits with code 1 if any exceeds the tolerance (default 0.05 kcal/mol).

Expected results

STRUCTURE (FILE)	LU COORDINATION	ΔE (KCAL/MOL)	ΔE WITHOUT D3 (KCAL/MOL)
`s3_2_10`	4N + 4O + H_2O (second sphere)	0.0	0.0
`s4_1_6`	4N + 4O + H_2O (second sphere)	4.2	4.2
`cn9_s4_1_6`	4N + 5O	6.9	7.8
`cn9_s3_2_10`	4N + 5O	14.9	15.9
`s5_0_6`	3N + 5O (one N dissociated)	18.9	19.4
`aq8_1` (reference)	8 H_2O + 1 H_2O (second sphere)	0.0	0.0
`aq9_0` (reference)	9 H_2O	1.2	1.8

File names: `sN_M_K` = start with N inner and M outer water molecules, conformer seed K (the final coordination can differ from the start); `cn9_*` = 4N + 5O cage; `aq*` = Lu^{3+} aqua ion.

Verification

On 9 November 2026 the package was installed in a new Python 3.13 virtual environment from `requirements.txt` and rerun (Linux x86_64, 2 cores):

- Steps 1–4 (48 conformers, CN 9 cages, aqua-ion reference): all GFN2-xTB energies identical to the deposited structures; `s3_2_10` again ranks lowest (`results/verification/xtb_ranking.json`).
- Step 5 for `aq8_1`, `aq9_0` and the newly generated `s3_2_10`: total energies identical to the deposited values to 10^{-8} Eh (`results/verification/compare_output.txt`).

Notes on reproducibility

- The deposited `cn9_*.xyz` files are the restrained geometries used for DFT. A new run of step 3 writes them as `cn9_*_restrained.xyz` (identical meaning) and also the released geometries `cn9_*.xyz`; release changes the GFN2-xTB energy by < 0.1 kcal/mol and keeps CN 9.
- Starting structures use fixed seeds, so with the versions in `requirements.txt` the conformer search gives the same structures. Other versions or platforms can change the numerics slightly, which may change which

conformer ranks lowest; check `work/xtb_ranking.json` before step 5. Route B does not depend on this and reproduces the DFT energies directly.

- Energies are electronic energies from single points on GFN2-xTB geometries with a double-zeta basis, without thermal or entropic corrections. GFN2-xTB is used for geometries only: in the aqua-ion test it wrongly favours CN 9 (by 8.8 kcal/mol), whereas the DFT level reproduces the experimental CN 8.
- Dipole moments of a cation depend on the origin; both the centre of nuclear charge and the Lu position are reported.

MATLAB

`matlab/` needs no toolboxes (MATLAB R2020a or newer for figure export).

```
cd lu_cyclam_dft/matlab
T = lu_dft_results;      % table of the deposited results
lu_dft_figures           % Figs. S1 and S2 (PNG 300 dpi and PDF) in ../figures
lu_dft_setup             % once: Python environment (Windows: in WSL2)
T = lu_dft_run('SkipXtb', true, 'Threads', 4);   % route B from MATLAB
```

On Windows, `lu_dft_setup` and `lu_dft_run` need WSL2 with Ubuntu. One-time setup: in PowerShell as administrator `wsl --install -d Ubuntu`, restart, then in Ubuntu `sudo apt update && sudo apt install -y python3-venv python3-pip`. `lu_dft_results` and `lu_dft_figures` work without Python or WSL.

Software and methods to cite

GFN2-xTB: Bannwarth, Ehlert, Grimme, J. Chem. Theory Comput. 15 (2019) 1652 · ALPB: Ehlert et al., J. Chem. Theory Comput. 17 (2021) 4250 · PBE0: Adamo, Barone, J. Chem. Phys. 110 (1999) 6158 · D3 / D3(BJ): Grimme et al., J. Chem. Phys. 132 (2010) 154104; J. Comput. Chem. 32 (2011) 1456 · def2: Weigend, Ahlrichs, Phys. Chem. Chem. Phys. 7 (2005) 3297; lanthanides: Gulde, Pollak, Weigend, J. Chem. Theory Comput. 8 (2012) 4062 · ECP28MWB: Dolg, Stoll, Preuss, J. Chem. Phys. 90 (1989) 1730 · ddCOSMO: Lipparini et al., J. Chem. Theory Comput. 9 (2013) 3637 · PySCF: Sun et al., J. Chem. Phys. 153 (2020) 024109 · tblite, simple-dftd3, ASE, RDKit, Basis Set Exchange (versions in `requirements.txt`).

Lu(III)–cyclam · Supporting Information, Section S1 · lu_cyclam_dft package
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